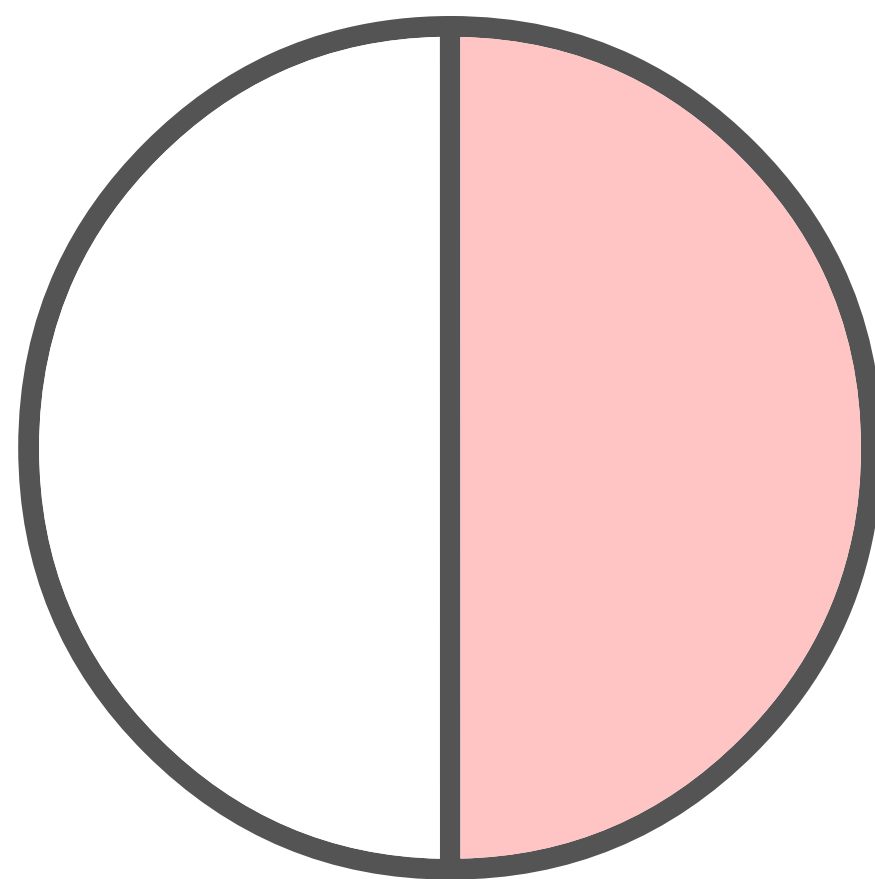


The background is a light teal color with various abstract patterns. In the top left, there is a pink triangle with a white dotted pattern. In the top right, there are several overlapping circles with pink outlines and some with pink dotted interiors. In the bottom left, there are pink and white concentric circles. In the bottom right, there is a pink wavy line pattern. A large, white, irregular blob shape is in the center, containing the text.

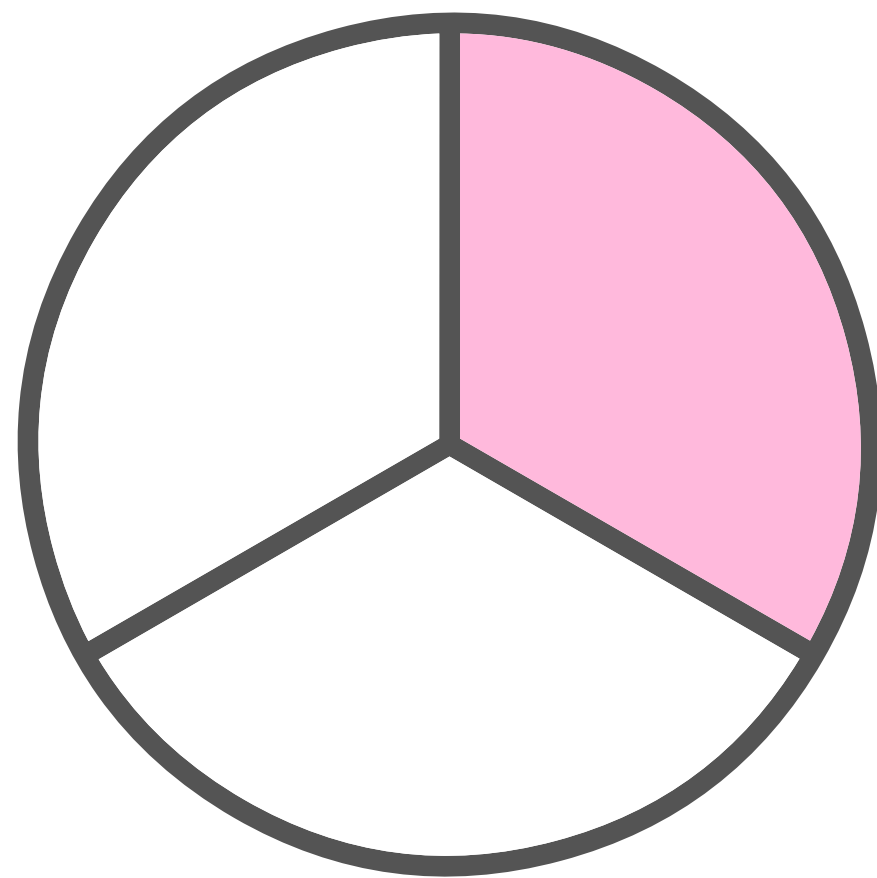
FUNKY FRACTIONS

ONE HALF

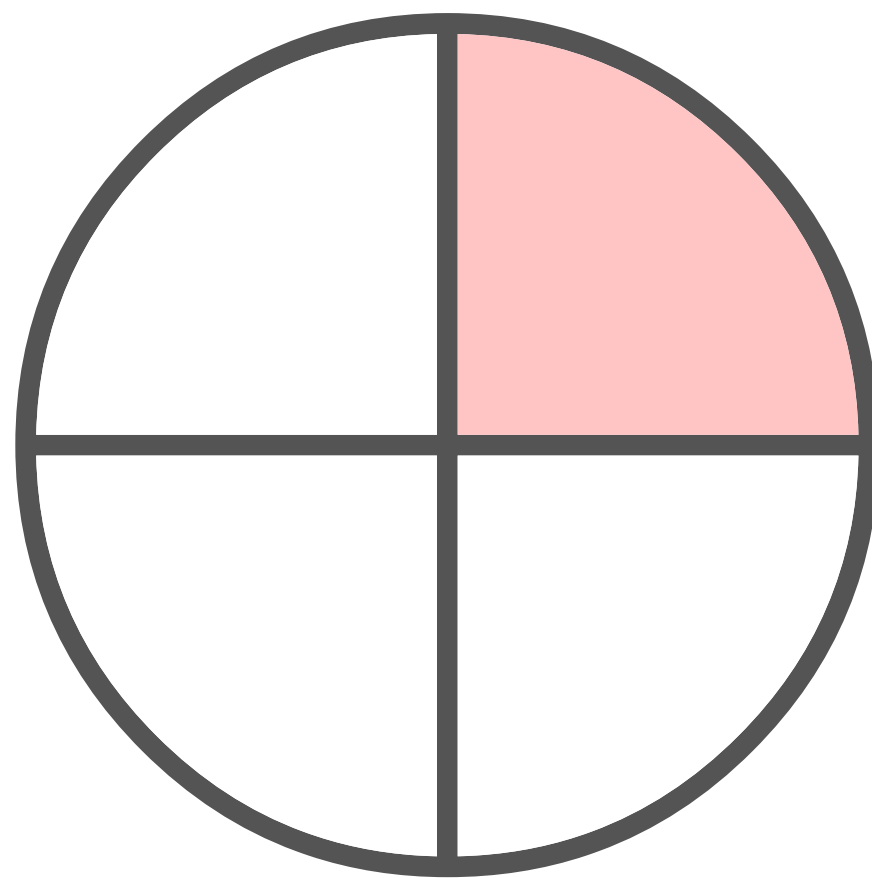
$1\frac{1}{2}$



ONE THIRD

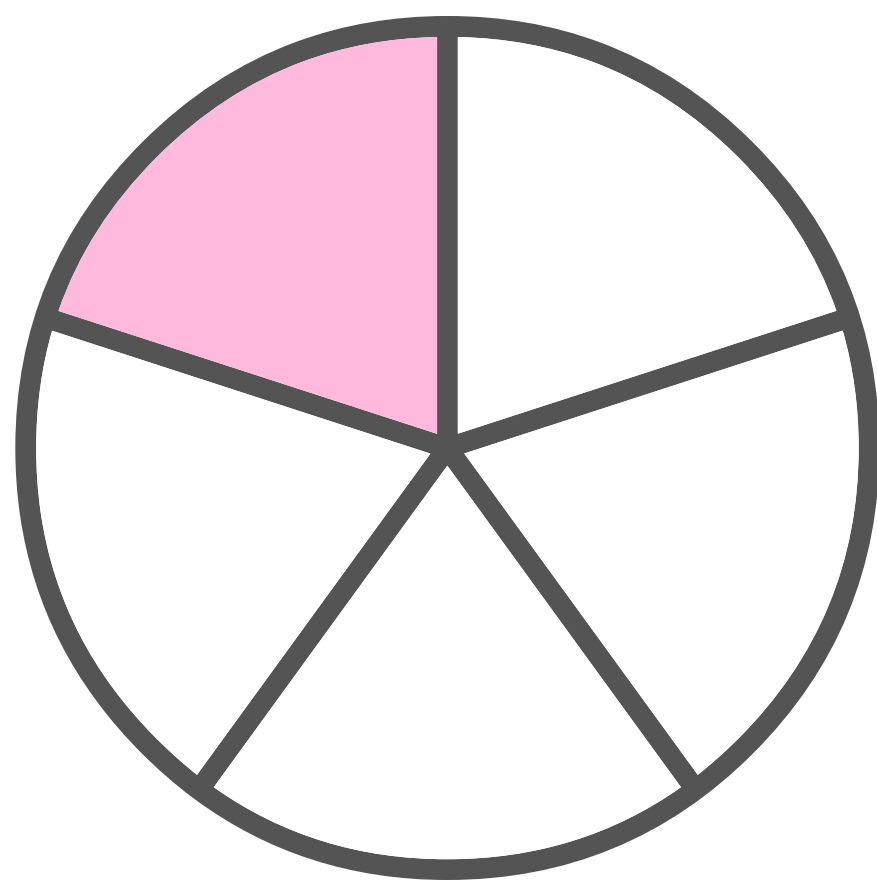
$$\frac{1}{3}$$


ONE FOURTH

$$\frac{1}{4}$$


ONE FIFTH

1
5

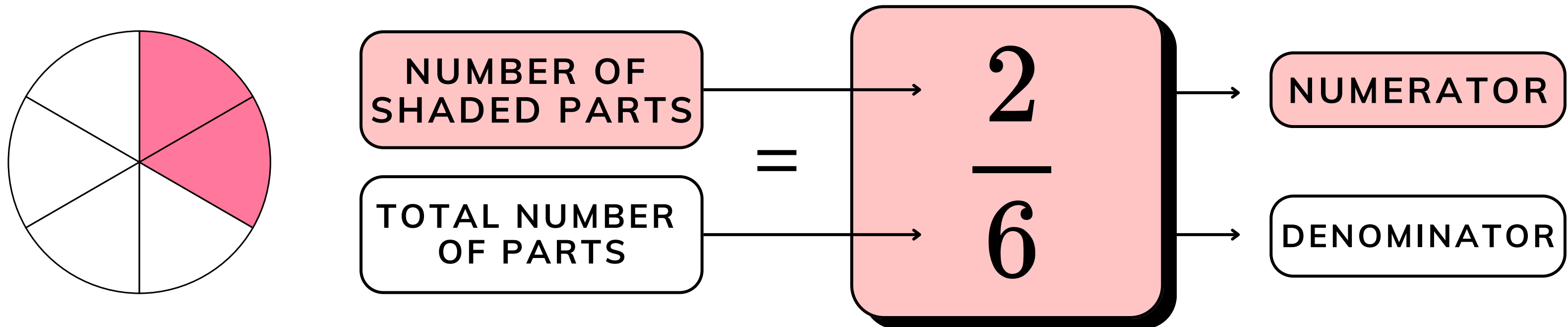


EQUIVALENT FRACTIONS



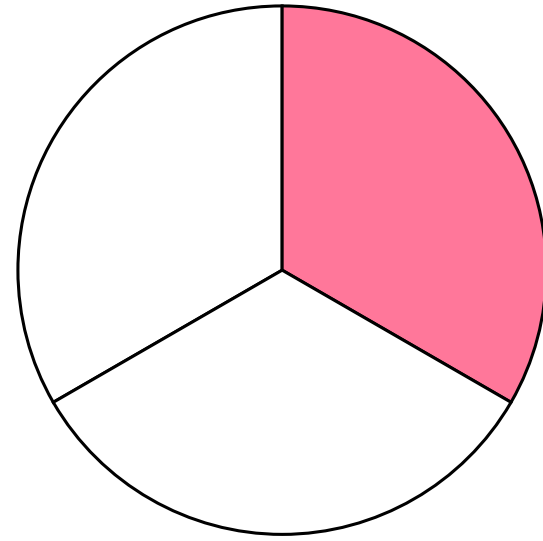
Writing Fractions from Area Models

Area models show fractions as parts of a whole.

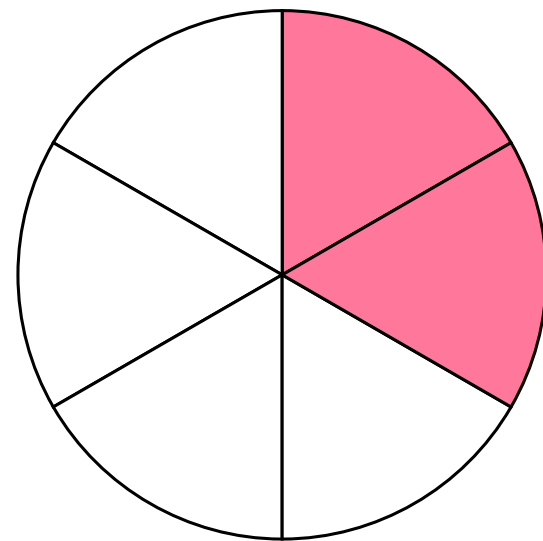


What are equivalent fractions?

Fractions that have the same value but with different numerators and denominators



=



$\frac{1}{3}$

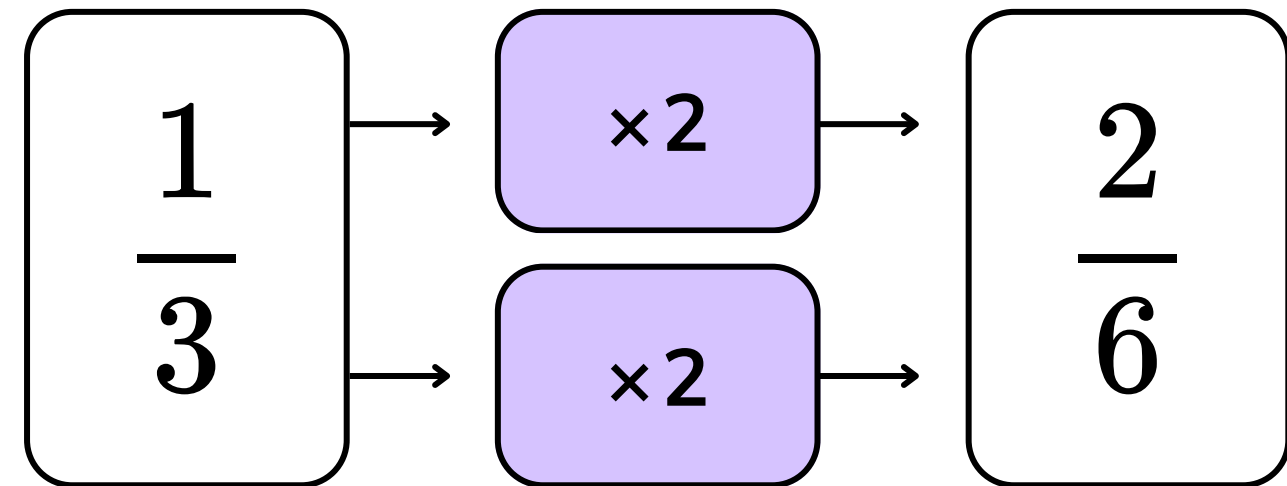
$\frac{2}{6}$

THEY
REPRESENT
THE SAME
PORTION OF
A WHOLE

Finding Equivalent Fractions

To find equivalent fractions,
multiply or divide both the
numerator and denominator
by the same number.

EXAMPLE 1

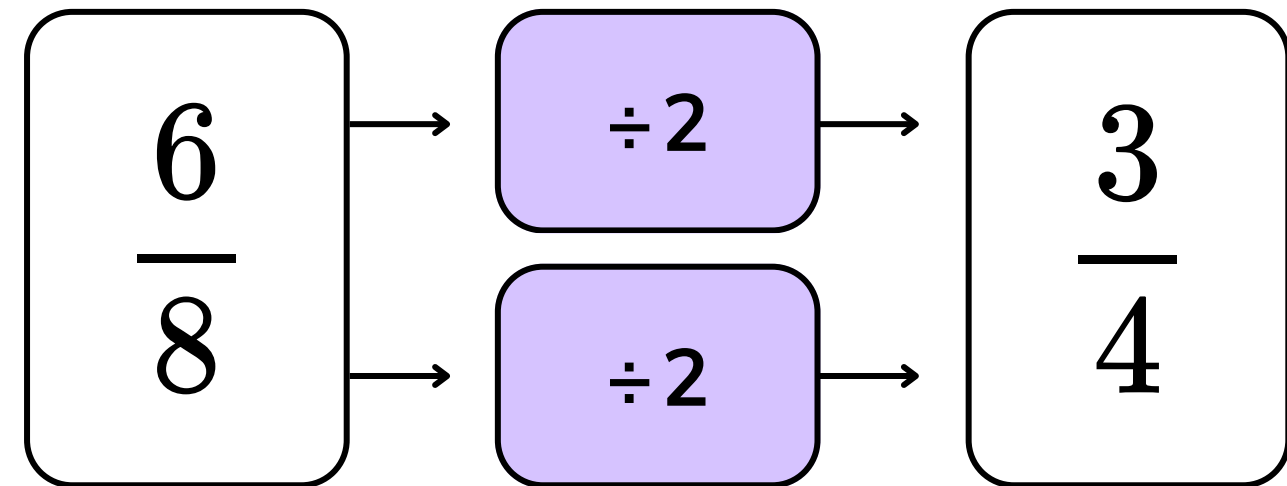


$\frac{1}{3}$ AND $\frac{2}{6}$ ARE
EQUIVALENT FRACTIONS

Finding Equivalent Fractions

To find equivalent fractions,
multiply or **divide** both the
numerator and denominator
by the same number.

EXAMPLE 2



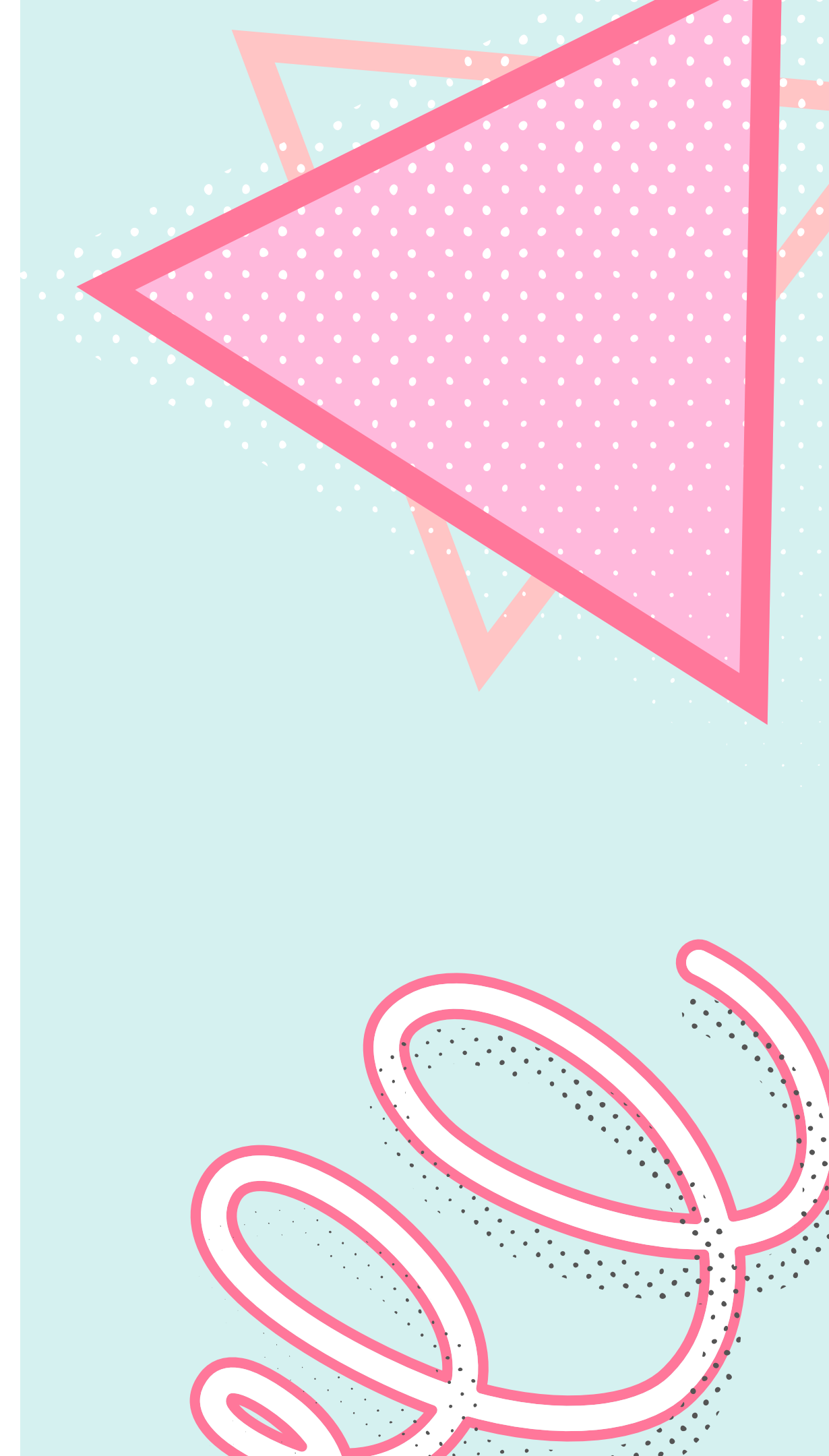
$\frac{6}{8}$ AND $\frac{3}{4}$ ARE
EQUIVALENT FRACTIONS



QUIZ TIME

QUESTION #1

Which is more, $\frac{1}{2}$ of a pizza or $\frac{1}{3}$ of a pizza?



QUESTION #2

True or False?

If I take $\frac{1}{2}$ of a cookie
there will be $\frac{1}{2}$ left.



QUESTION #3

True or False?

$\frac{1}{5}$ of a cake is less
than $\frac{1}{3}$ of a cake.





**GREAT
JOB!**

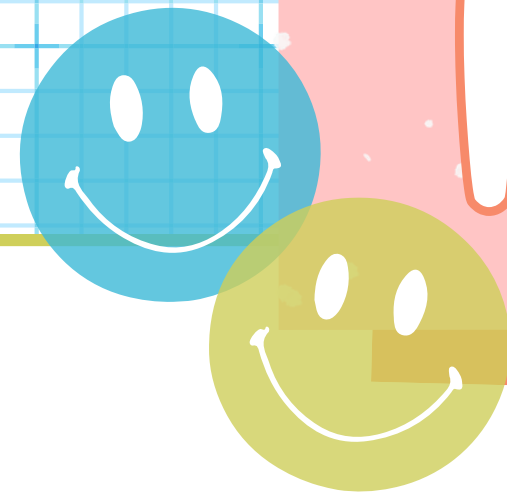
Identifying



Fractions on a

Number Line

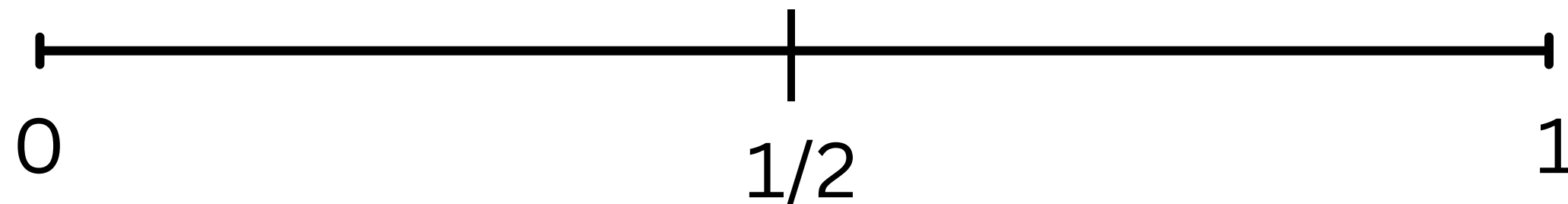


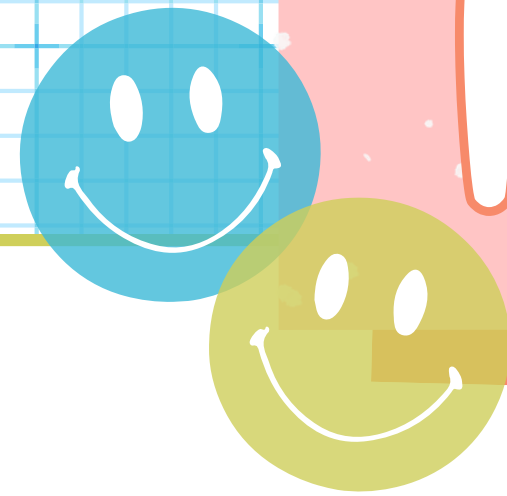


Introduction

Fractions can be placed on a number line the same way that you would place numbers.

The fraction $\frac{1}{2}$ is in between 0 and 1. The distances on either side of $\frac{1}{2}$ are equal.

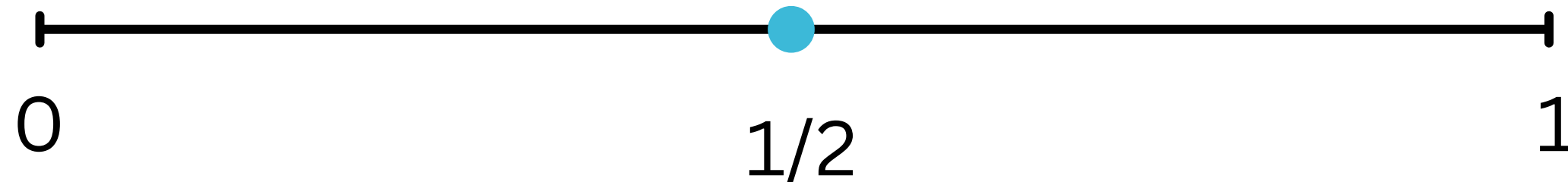




Introduction

If you look at the number line closely, you will see that the number line is now divided into two equal parts.

1 is the whole. $\frac{1}{2}$ is the part.

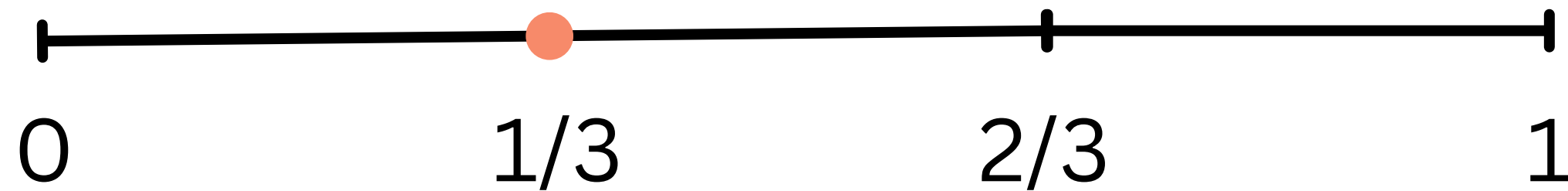




Let's Try!

Let's think about the fraction $\frac{1}{3}$.

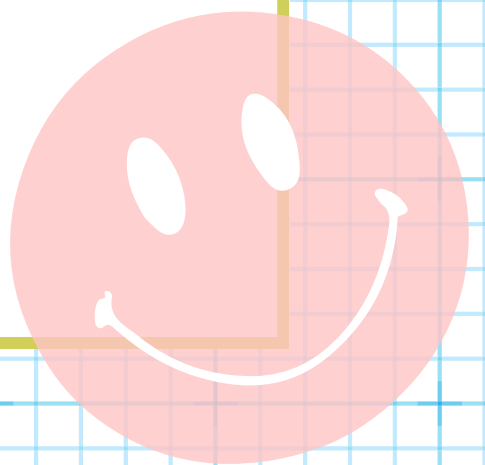
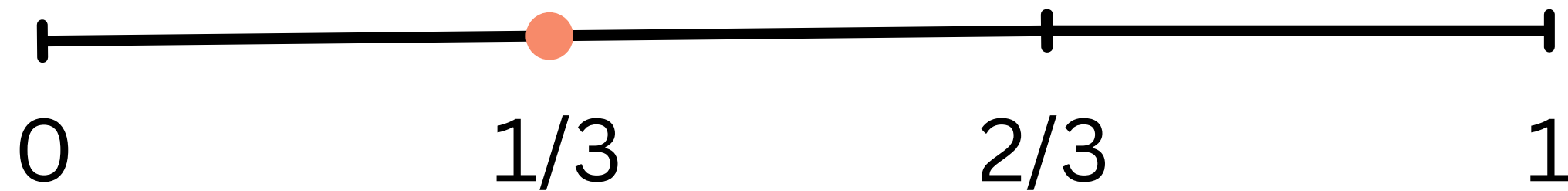
1 is the whole. There are 3 parts in the 1 whole.





Let's Try!

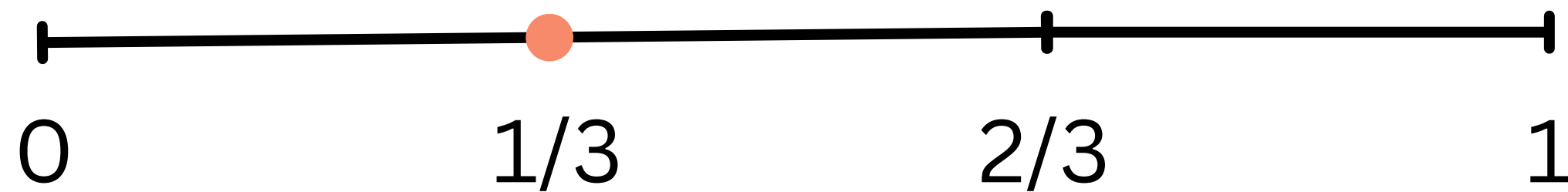
What would be another way to write 1 on this number line?





Let's Try!

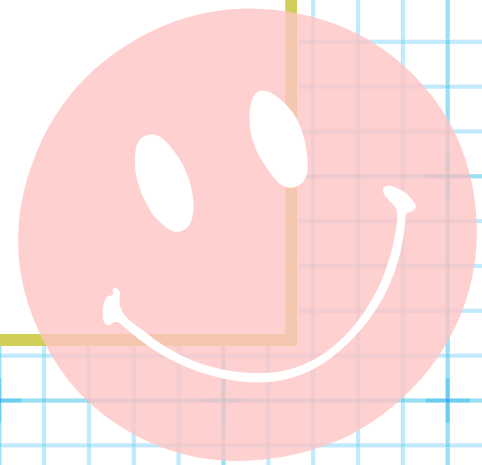
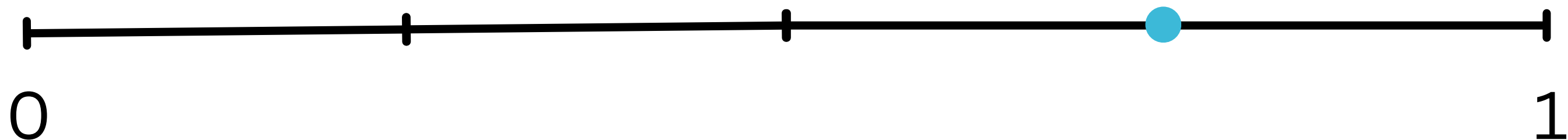
$$1 = 3/3$$





Let's Try!

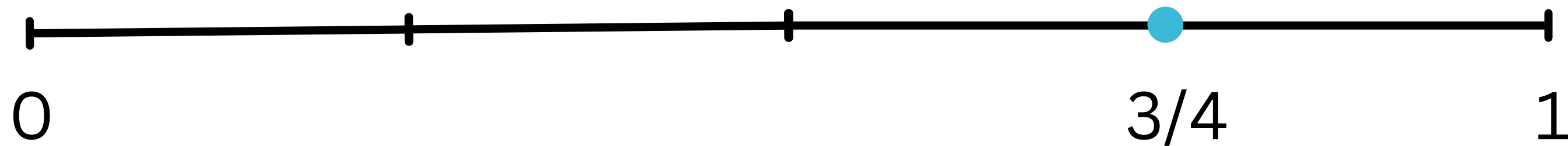
What fraction is marked on the number line?





Let's Try!

It is $\frac{3}{4}$! The number line is divided into 4 equal sections between 0 and 1.



COMPARING



FRACTIONS

**MAKING DENOMINATORS
THE SAME HELPS US
COMPARE FRACTIONS!**

WHAT IS A COMMON DENOMINATOR?



$$\frac{5}{8}$$

$$\frac{3}{4}$$

These fractions do not have the same denominator.

BUT - we can use equivalent fractions to make these fractions have common denominators.

**WE CAN MAKE THIS FRACTION
HAVE A DENOMINATOR OF 8**

$$\frac{3}{4}$$

The diagram illustrates the process of finding an equivalent fraction with a denominator of 8. It shows the fraction $\frac{3}{4}$ on the left, followed by an equals sign, and then the fraction $\frac{6}{8}$ on the right. A blue curved arrow points from the numerator 3 to the numerator 6, with the label 'x2' above it. Another blue curved arrow points from the denominator 4 to the denominator 8, with the label 'x2' below it.

$$\frac{3}{4} = \frac{6}{8}$$

**THESE FRACTIONS
ARE EQUIVALENT**

$$\frac{5}{8}$$

$$\frac{3}{4}$$



$$\frac{6}{8}$$

**NOW WE CAN EASILY SEE
WHICH FRACTION IS BIGGER!**

$$\frac{5}{8}$$



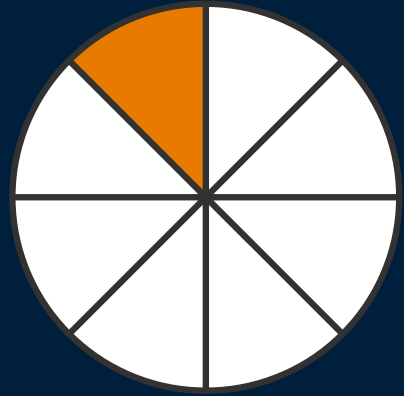
$$\frac{3}{4}$$



$$\frac{6}{8}$$

***First up,
addition!***





When adding fractions with common denominators, we ADD the numerators.

$$\frac{1}{8} + \frac{4}{8} = \frac{5}{8}$$

If our answer is not in simplest form, we must simplify it.

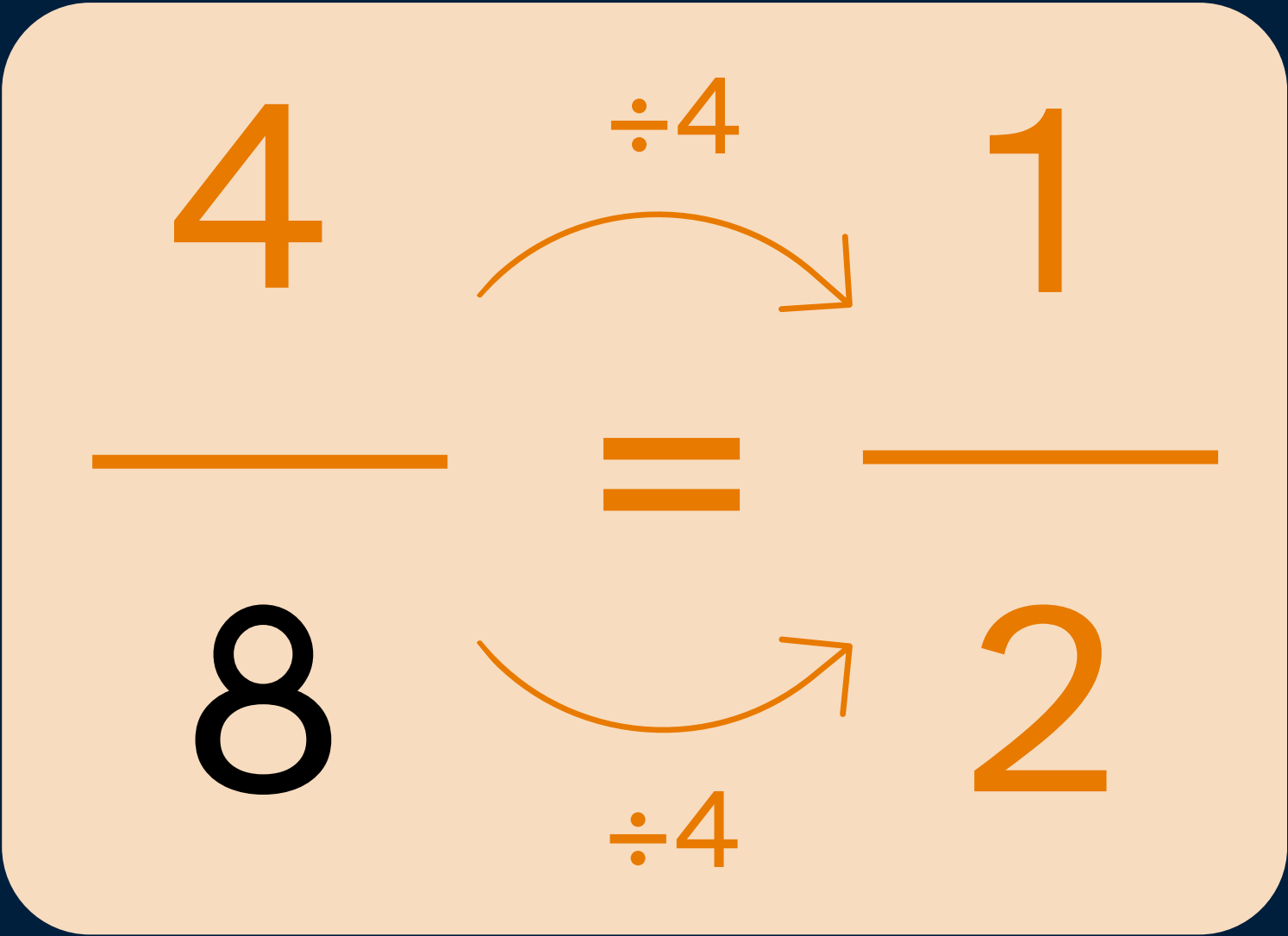


$$\frac{1}{8} + \frac{3}{8} = \frac{4}{8}$$



$$\frac{4}{8} = \frac{1}{2}$$

Our working out would look like this:

$$\frac{1}{8} + \frac{3}{8} = \frac{4}{8} = \frac{1}{2}$$


The diagram illustrates the simplification of the fraction $\frac{4}{8}$ to $\frac{1}{2}$. It shows the numerator 4 being divided by 4 to become 1, and the denominator 8 being divided by 4 to become 2. The division is indicated by arrows and the label $\div 4$.



***Here's what we
do when our
denominators
do not match!***

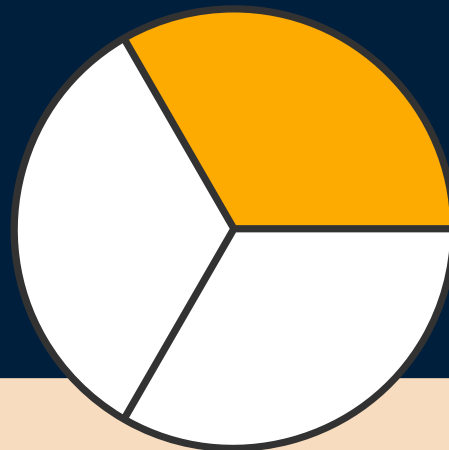


We can create **EQUIVALENT FRACTIONS**.

We must find the **Lowest Common Denominator (LCD)** for our denominators.

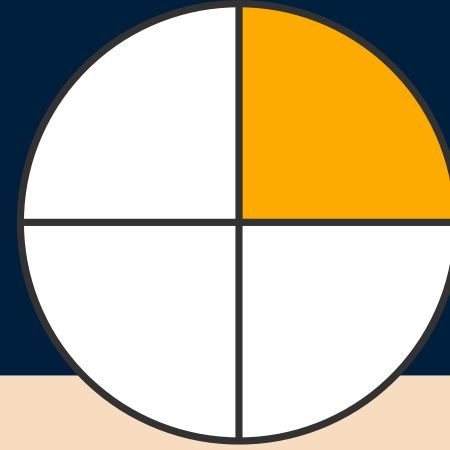
$$\frac{1}{3} + \frac{1}{4} = ?$$

**The Lowest Common Denominator
(LCD) for 3 and 4 is 12.**



$$\frac{1}{3} \overset{x4}{\curvearrowright} \frac{4}{12}$$

$\frac{1}{3} = \frac{4}{12}$



$$\frac{1}{4} \overset{x3}{\curvearrowright} \frac{3}{12}$$

$\frac{1}{4} = \frac{3}{12}$

Now, we can add our fractions.

$$\frac{4}{12} + \frac{3}{12} = \frac{7}{12}$$

***Now, let's do
subtraction!***



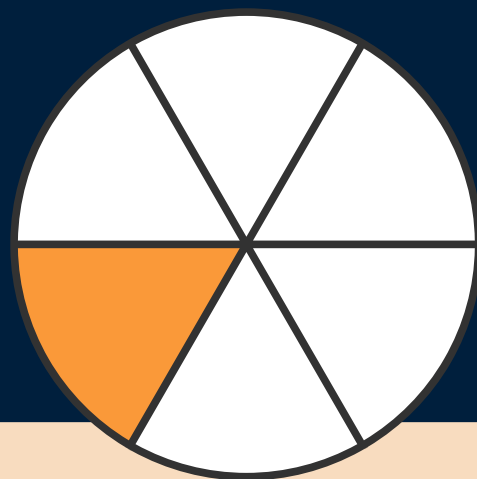


These fractions have **common denominators**,
so we **SUBTRACT** the numerators.

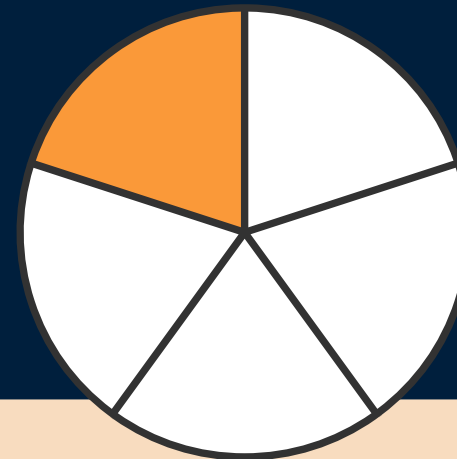
$$\frac{5}{7} - \frac{2}{7} = \frac{3}{7}$$

The **Lowest Common Denominator (LCD)**
for 6 and 5 is 30.

$$\frac{5}{6} - \frac{1}{5} = ?$$



$$\frac{5}{6} \overset{x5}{\curvearrowright} \frac{25}{30} = \frac{5}{6} \overset{x5}{\curvearrowright} \frac{25}{30}$$



$$\frac{1}{5} \overset{x6}{\curvearrowright} \frac{6}{30} = \frac{1}{5} \overset{x6}{\curvearrowright} \frac{6}{30}$$

Now, we can subtract our fractions.

$$\frac{25}{\boxed{30}} - \frac{6}{\boxed{30}} = \frac{19}{\boxed{30}}$$

Remember:

When the denominators are the same, we add or subtract the numerators only.

If the denominators are not the same, we can use equivalent fractions.



THANK YOU!

